Chapter 2 Checklist

Polynomial and Rational Functions

Ron Larson and Robert P. Hostetler, *Precalculus with Limits*, 2nd ed., p. 125. Tick each box as you go.

Work through the four parts in order: read each section, practise the matching Duolingo units, work the book-only topics that Duolingo does not cover, then finish the chapter. Duolingo units are grouped by lane (a grade or a topic in the app) and listed in the order the app shows them, so you can follow one lane from top to bottom. **full** = the unit practises the section's main skill, **partial** = part of it, **prereq** = a warm-up from an earlier grade.

In this chapter: Grades 11 and 12 between them cover polynomial, complex, and rational functions thoroughly. Duolingo has no unit on polynomial long or synthetic division (2.3) or on nonlinear inequalities (2.7); use the book for those.

1. Read the textbook sections

- ☐ 2.1 Quadratic Functions and Models — p. 126
- ☐ 2.2 Polynomial Functions of Higher Degree — p. 136
- ☐ 2.3 Polynomial and Synthetic Division — p. 150
- ☐ 2.4 Complex Numbers — p. 159
- ☐ 2.5 Zeros of Polynomial Functions — p. 166
- ☐ 2.6 Rational Functions — p. 181
- ☐ 2.7 Nonlinear Inequalities — p. 194

2. Practise on Duolingo

Pick one lane and work down it; the grade lanes go deepest. The note after each unit gives its rating and the textbook sections it serves.

Grade 9

- ☐ Solve one-variable inequaltiies *prereq, 2.7* — unit 11
- ☐ Compound 'or' inequalities *prereq, 2.7* — unit 12
- ☐ Compound 'and' inequalities *prereq, 2.7* — unit 13
- ☐ Tables of quadratic functions *full, 2.1* — unit 56
- ☐ Graphs of quadratic functions *full, 2.1* — unit 57
- ☐ Equations of quadratic functions *full, 2.1* — unit 58
- ☐ Zeros of quadratic functions *full, 2.1* — unit 59
- ☐ Factored form *full, 2.1* — unit 60
- ☐ Vertex *full, 2.1* — unit 61
- ☐ Vertex form *full, 2.1* — unit 62

Grade 11

- ☐ Intro to polynomial functions *full, 2.2* unit 12
- ☐ Relative maxima and minima of polynomial functions *full, 2.2* unit 13
- ☐ Zeros of polynomial functions *full, 2.2, 2.5* unit 14
- ☐ Multiplicity of polynomial functions *full, 2.2* unit 15
- ☐ End behavior of odd polynomial functions *full, 2.2* unit 16
- ☐ End behavior of even polynomial functions *full, 2.2* unit 17
- ☐ The imaginary number, i *full, 2.4* unit 18
- ☐ Square roots of negative numbers *full, 2.4* unit 19
- ☐ Patterns of exponentiating i *full, 2.4* unit 20
- ☐ Multiply complex numbers *full, 2.4* unit 21
- ☐ Complex roots *full, 2.5* unit 22
- ☐ Complex conjugates *full, 2.4* unit 23
- ☐ Intro to rational functions *full, 2.6* unit 24
- ☐ Y-intercept of rational functions *full, 2.6* unit 25
- ☐ X-intercepts of rational functions *full, 2.6* unit 26
- ☐ Vertical asymptotes of rational functions *full, 2.6* unit 27
- ☐ Rewrite rational functions *full, 2.6* unit 28
- ☐ End behavior approaching a horizontal line *full, 2.6* unit 29
- ☐ End behavior approaching y=0 *full, 2.6* unit 30
- ☐ End behavior approaching slant asymptotes *full, 2.6* unit 31

Grade 12

- ☐ Polynomial function graphs *full, 2.2* unit 2
- ☐ Roots of polynomial functions *full, 2.5* unit 3
- ☐ Rational root theorem *full, 2.5* unit 4
- ☐ Polynomial function equations from graphs *full, 2.2* unit 5
- ☐ Vieta's formula for quadratic polynomials *partial, 2.5* unit 6
- ☐ Vieta's formula for cubic polynomials *partial, 2.5* unit 7
- ☐ Zeros of rational functions *full, 2.6* unit 8
- ☐ Vertical asymptotes of rational functions *full, 2.6* unit 9
- ☐ Horizontal asymptotes of rational functions *full, 2.6* unit 10
- ☐ Holes in rational functions *full, 2.6* unit 11
- ☐ Slant asymptotes of rational functions *full, 2.6* unit 12
- ☐ Intermediate value theorem *partial, 2.2* unit 48

Algebraic Graphing

- ☐ Intro to quadratic functions *full, 2.1* unit 11
- ☐ Quadratic graphs and equations *full, 2.1* unit 12
- ☐ Factored form of quadratics *full, 2.1* unit 13
- ☐ Vertex form of quadratics *full, 2.1* unit 14
- ☐ Polynomial function equations *full, 2.2* unit 30
- ☐ Polynomial function graphs *full, 2.2* unit 31
- ☐ Polynomial graphs and roots *full, 2.2* unit 32
- ☐ Rational function equations and graphs *full, 2.6* unit 33
- ☐ Rewrite and graph rational functions *full, 2.6* unit 34
- ☐ Rational function asymptotes *full, 2.6* unit 35
- ☐ Holes and slant asymptotes *full, 2.6* unit 36

3. Section by section

For each section: what Duolingo covers (the units are ticked off in part 2) and the book-only topics to work in the textbook. Tick each book-only topic once you have worked examples of it.

2.1 Quadratic Functions and Models p. 126

Thoroughly covered in grade 9 and the Algebraic Graphing topic. Completing the square to reach vertex form is lighter in the app.

Duolingo: **Grade 9:** 56. Tables of quadratic functions (full); 57. Graphs of quadratic functions (full); 58. Equations of quadratic functions (full); 59. Zeros of quadratic functions (full); 60. Factored form (full); 61. Vertex (full); 62. Vertex form (full)
Algebraic Graphing: 11. Intro to quadratic functions (full); 12. Quadratic graphs and equations (full); 13. Factored form of quadratics (full); 14. Vertex form of quadratics (full)

Book only:

- ☐ Completing the square to write vertex form
- ☐ Maximum and minimum word problems

2.2 Polynomial Functions of Higher Degree p. 136

Grade 11 covers end behaviour, zeros, multiplicity, and relative extrema; grade 12 adds writing equations from graphs. The Leading Coefficient Test is taught as end behaviour.

Duolingo: **Grade 11:** 12. Intro to polynomial functions (full); 13. Relative maxima and minima of polynomial functions (full); 14. Zeros of polynomial functions (full); 15. Multiplicity of polynomial functions (full); 16. End behavior of odd polynomial functions (full); 17. End behavior of even polynomial functions (full)
Grade 12: 2. Polynomial function graphs (full); 5. Polynomial function equations from graphs (full); 48. Intermediate value theorem (partial)
Algebraic Graphing: 30. Polynomial function equations (full); 31. Polynomial function graphs (full); 32. Polynomial graphs and roots (full)

Book only:

- ☐ Using the Intermediate Value Theorem to locate a zero

2.3 Polynomial and Synthetic Division p. 150

Duolingo has no unit on polynomial long division, synthetic division, or the Remainder and Factor Theorems. Use the book; grade 12's Rational root theorem assumes this skill.

Duolingo: no unit covers this section.

Book only:

- ☐ Polynomial long division
- ☐ Synthetic division
- ☐ The Remainder Theorem and the Factor Theorem

2.4 Complex Numbers p. 159

Grade 11 covers the imaginary unit, arithmetic, conjugates, and powers of i. Division by a complex number is practised through conjugates; the complex plane is book only.

Duolingo: **Grade 11:** 18. The imaginary number, i (full); 19. Square roots of negative numbers (full); 20. Patterns of exponentiating i (full); 21. Multiply complex numbers (full); 23. Complex conjugates (full)

Book only:

- ☐ Plotting complex numbers in the complex plane

2.5 Zeros of Polynomial Functions p. 166

Rational zero test, complex zeros, and conjugate pairs are covered across grades 11 and 12. Vieta's formulas go beyond the section. Descartes' Rule of Signs and bounds are book only.

Duolingo: **Grade 11:** 14. Zeros of polynomial functions (prereq); 22. Complex roots (full)
Grade 12: 3. Roots of polynomial functions (full); 4. Rational root theorem (full); 6. Vieta's formula for quadratic polynomials (partial); 7. Vieta's formula for cubic polynomials (partial)

Book only:

- ☐ The Fundamental Theorem of Algebra
- ☐ Descartes's Rule of Signs
- ☐ Upper and lower bounds for real zeros

2.6 Rational Functions p. 181

Very well covered. Grade 11 walks through intercepts and each kind of asymptote; grade 12 adds holes and slant asymptotes explicitly.

Duolingo: **Grade 11:** 24. Intro to rational functions (full); 25. Y-intercept of rational functions (full); 26. X-intercepts of rational functions (full); 27. Vertical asymptotes of rational functions (full); 28. Rewrite rational functions (full); 29. End behavior approaching a horizontal line (full); 30. End behavior approaching y=0 (full); 31. End behavior approaching slant asymptotes (full)
Grade 12: 8. Zeros of rational functions (full); 9. Vertical asymptotes of rational functions (full); 10. Horizontal asymptotes of rational functions (full); 11. Holes in rational functions (full); 12. Slant asymptotes of rational functions (full)
Algebraic Graphing: 33. Rational function equations and graphs (full); 34. Rewrite and graph rational functions (full); 35. Rational function asymptotes (full); 36. Holes and slant asymptotes (full)

Book only:

- ☐ Applications of rational functions

2.7 Nonlinear Inequalities p. 194

Duolingo has no unit on polynomial or rational inequalities or the sign-chart method. Grade 9 one-variable inequalities are the only warm-up.

Duolingo: **Grade 9:** 11. Solve one-variable inequaltiies (prereq); 12. Compound 'or' inequalities (prereq); 13. Compound 'and' inequalities (prereq)

Book only:

- ☐ Polynomial inequalities
- ☐ Rational inequalities
- ☐ The test-interval (sign chart) method

4. Finish the chapter

- ☐ Chapter Summary p. 204
- ☐ Review Exercises p. 206
- ☐ Chapter Test p. 210
- ☐ Proofs in Mathematics p. 211
- ☐ Problem Solving p. 213

Duolingo unit numbers are each unit's position in that grade's or topic's path in the Duolingo Math app, as recorded on 2026-10-07 (iOS); the app may reorder them. Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.